ONLINE ASSESSMENT SYSTEM USING AI-BASED MONITORING

ABSTRACT

The Online Assessment System using AI-Based Monitoring is an advanced platform designed to conduct online examinations in a secure, reliable, and automated manner. The system provides role-based access for administrators, faculty members, and students. Administrators can manage users, monitor system activities, and view examination records. Faculty members can create and schedule exams, generate questions manually or automatically, and analyze student performance. Students can register, view exam schedules, take assessments, and receive their results. All examination processes are automated, reducing manual effort and minimizing the need for physical examination centers. By providing real-time monitoring, identity verification, and intelligent alert mechanisms, the system ensures a fair, transparent, and unbiased assessment environment, making it highly suitable for remote learning and online education systems.

Keywords: Online Assessment, AI-Based Monitoring, Monitoring System, Face Recognition, Academic Integrity, Real-Time Surveillance, Identity Verification, Automated Examination, Secure Online Exams.

STATEMENT ABOUT THE PROBLEM

With the rapid growth of online education, conducting secure and trustworthy online examinations has become a major challenge. Traditional examination methods rely on physical presence and manual invigilation, which are impractical in remote environments. Existing online exam systems often lack effective monitoring and identity verification mechanisms, making them vulnerable to malpractice such as cheating, impersonation, and unauthorized collaboration. In many cases, students can access external resources, receive help from others, or substitute another person to take the exam on their behalf. This lack of proper supervision reduces the credibility of online assessments and affects the fairness of the evaluation process. Hence, there is a strong need for an intelligent, automated online assessment system that can authenticate the candidate and monitor their behavior during the examination. The system should be capable of detecting suspicious activities in real time and provide a secure virtual exam environment without the requirement of physical examination centers.

WHY IS THE PARTICULAR TOPIC CHOSEN?

The topic "Online Assessment System Using AI-Based Monitoring" has been chosen because maintaining academic integrity in online examinations is a major concern for educational institutions worldwide. As digital learning continues to grow, there is an increasing demand for secure, scalable, and reliable online assessment solutions. By integrating AI technologies such as face recognition, behavior analysis, and object detection, the proposed system helps in reducing malpractice and ensures that exams are conducted in a fair and transparent manner. This topic is highly relevant in today's education system and offers practical solutions to real-world problems faced by educators and institutions in the remote learning environment.

SCOPE:

The scope of this project involves the design and development of an AI-based online assessment and monitoring system that supports secure remote examinations. The system will include:

Secure user registration and login for students, faculty, and administrators

Real-time identity verification using facial recognition

Continuous AI monitoring to detect suspicious activities (multiple faces, mobile phone usage, abnormal behavior)

Automated exam scheduling and question management

Online test interface with real-time surveillance

Result generation and performance analysis

The system is intended for educational institutions, training centers, and corporate organizations that conduct online assessments and certification exams.

OBJECTIVE OF THE PROJECT:

The main objective of this project is to develop a secure and intelligent online assessment platform using AI-based monitoring techniques. The system aims to:

Prevent exam malpractice such as cheating and impersonation.

Verify student identity using facial recognition

Monitor candidate behavior throughout the exam session

Detect and alert suspicious activities in real time

Automate exam scheduling, execution, and evaluation

Provide a user-friendly and scalable solution for remote assessments

Ultimately, the goal is to enhance the credibility and reliability of online examinations.

EXISTING METHOD

In the existing scenario, most online examination systems provide only basic features such as question paper creation, online test conduction, and automatic result generation. These platforms mainly depend on time restrictions and basic webcam monitoring, which are not sufficient to prevent cheating. There is no advanced behavior analysis or effective candidate authentication in many systems. As a result, students can easily bypass security measures by using external devices, switching browser tabs, or getting help from others, making current online assessments less reliable.

DISADVANTAGES

1. Lack of Real-time Monitoring: Many current systems fail to provide real-time monitoring, making it difficult to detect cheating or suspicious behaviors during the exam.

2. No Identity Verification: Most platforms do not have advanced identity verification mechanisms like face recognition, allowing impersonation or unauthorized users to take exams.

3. Vulnerable to Cheating: Without effective monitoring, students can easily use unfair means such as accessing unauthorized resources, using mobile phones, or receiving assistance during the exam.

4. Manual Proctoring Dependency: Some systems rely on manual invigilation via video conferencing, which is time-consuming, labour-intensive, and prone to human error.

5. Limited Scalability: Existing platforms often struggle to scale for large numbers of students without compromising on exam integrity, making them unsuitable for larger institutions.

6. Inefficient Exam Scheduling: Lack of automation in creating and scheduling exams can result in administrative overhead for instructors, making the process cumbersome and inefficient.

7. No Proactive Alerts: Instructors are usually not notified of suspicious activities in real-time, limiting their ability to intervene promptly to prevent cheating.

PROPOSED SYSTEM:

The proposed Online Assessment System Using AI-Based Monitoring overcomes the limitations of existing systems by implementing intelligent monitoring techniques. It continuously verifies the student’s identity and tracks behavior through the webcam using AI models.

The system is designed to detect:

Multiple persons in front of the camera

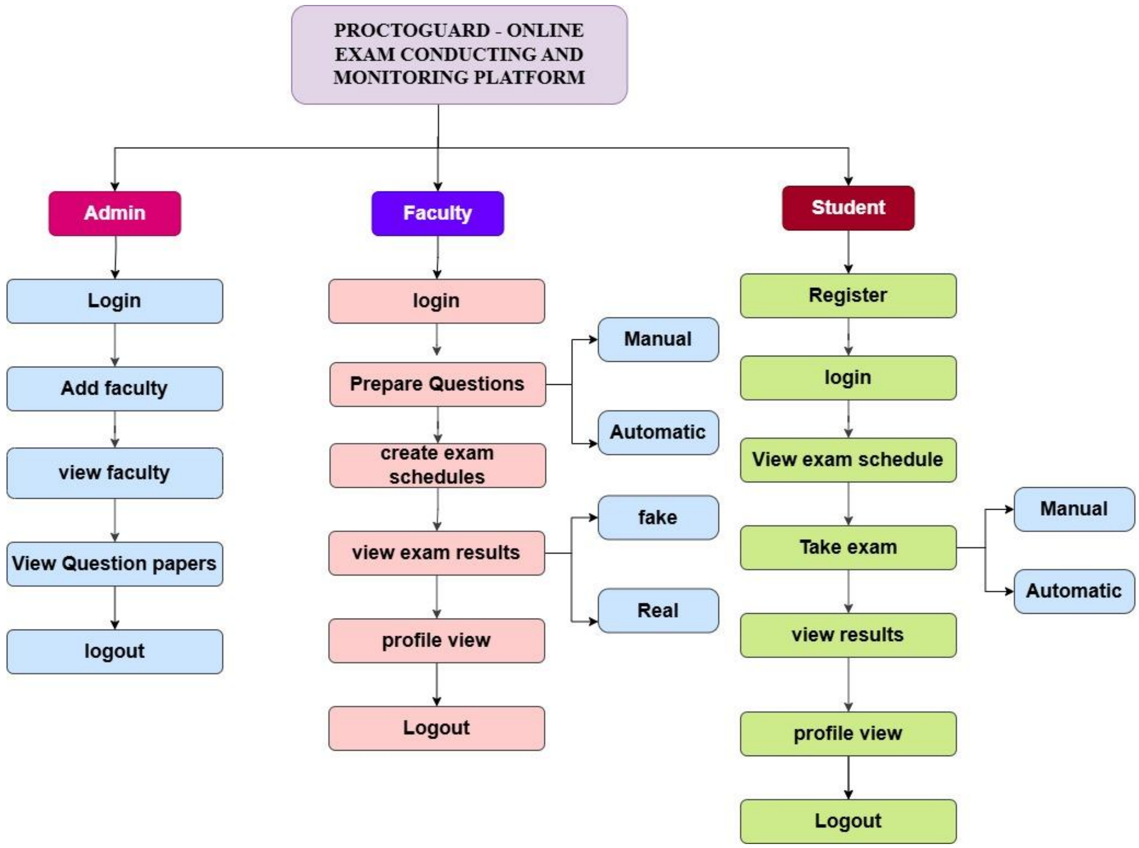
Usage of mobile phones or electronic devices

Irregular head movements or looking away frequently

Absence from the camera for extended periods

It includes automated exam scheduling, a secure login system, customized alerts for suspicious activities, and real-time monitoring to maintain exam integrity. The platform provides a safe, reliable, and scalable environment for conducting remote examinations without the need for physical exam centers.

PROJECT FLOW:



ADVANTAGES:

1. Ensures high academic integrity.
2. Uses AI-based real-time monitoring
3. Prevents impersonation and cheating
4. Fully automated examination process
5. Scalable for large number of students
6. Saves time and resources
7. Suitable for remote learning environments
8. Provides accurate and reliable results

SOFTWARE FRONT END REQUIREMENTS

H/W CONFIGURATION:

- Processor - I3/Intel Processor
- Hard Disk - 160GB
- Key Board - Standard Windows Keyboard
- Mouse - Two or Three Button Mouse
- Monitor - SVGA
- RAM - 8GB

S/W CONFIGURATION:

- Operating System : Windows 7/8/10
- Server side Script : HTML, CSS, Bootstrap & JS
- Programming Language : Python
- Libraries : Django, Pandas, Mysql.connector, Os, Smtplib, Numpy
- IDE/Workbench : VS Code
- Technology : Python 3.6+
- Server Deployment : Xampp Server
- Database : MySQL

MODULES/IMPLEMENTATION

The system is divided into three main modules:

1.**Admin Module**

Login: The administrator logs in using default credentials (email and password).

Add Faculty: The admin adds faculty members by entering their details. Login credentials are sent to the faculty via email.

Add Students: The admin views and manages registered student details.

View Question Papers: The admin can view and monitor the question papers created by faculty.

Logout: The admin can securely log out from the system.

2. **Faculty Module**

Login: Faculty members log in using the credentials provided by the admin.

Prepare Questions: Faculty can create questions manually or automatically using an AI-based model.

Create Exam Schedule: After adding questions, faculty can schedule the exam for students.

View Exam Results: Once the exam is completed, faculty can view the results of the students.

If proper face detection is found and no mobile phone is detected, the attempt is marked as Real and the result email is sent to the student.

If face detection fails or a mobile phone is detected, the attempt is marked as Fake.

Profile Update: Faculty can view and update their profile details.

Logout: Faculty can securely log out from the system.

3. **Student Module**

Registration: Students register by entering their personal details. During registration, face data is captured for future identity verification.

Login: Students can log in using their registered credentials.

View Exam Schedule: Students can view their scheduled exams provided by the faculty.

Attend Exam: During the exam, the webcam is activated: Face recognition verifies identity, Mobile phone detection is monitored, Behavior is continuously tracked using AI.

View Result: After completion, students can view their result if their attempt is marked as Real.

Profile Update: Students can update their personal information.

Logout: Students can securely log out from the system.